Selective Language Damage with Parameter-Decomposition Subcomponents

Sergei Solovev

Independent researcher

Version 0.1.1 · 15 October 2026

Changing one language while sparing a related language tests how selectively a model can be edited. In a four-layer, 67M-parameter model, adVersarial Parameter Decomposition (VPD) approximates weight matrices with sums of rank-one subcomponents. A fixed rule selects one subcomponent per target language using target and English text. The edits change its coefficient while leaving the decomposition residual fixed. I compare six edits for German, Spanish and French with rank-one LoRA, full-matrix fine-tuning and neuron edits. All methods are calibrated to raise target cross-entropy by 0.5 or 1 nat/token on validation text.

On 58 new European Parliament session dates, the subcomponent edits raised cross-entropy in each target's related language (Dutch, Portuguese and Italian) by 0.0003–0.0053 nat/token, compared with 0.027–0.876 across comparator runs. The learned updates, trained to preserve English, preserved English predictions better. In an exploratory check on earlier development dates, an expanded training recipe generally reduced the learned updates' related-language damage, but that damage stayed above the subcomponent edits'. Whether this advantage depends on the VPD dictionary (the decomposition's set of subcomponents) and how the selection rule performs on new tasks remain unresolved.

1 Introduction

A model edit that changes one behaviour usually changes others too. Model-editing work evaluates *locality*: inputs outside an edit's scope should keep their predictions (Yao et al. 2023, sec. 2). ROME, for example, scores an edited fact by whether the model retains facts about nearby subjects (Meng et al. 2022, sec. 3.3). For a language edit, a demanding test is a related language. Zeroing parameters important for Spanish also raises Italian and Portuguese perplexity (Zhang et al. 2024, Table 11), and deactivating French neurons affects Spanish (Tang et al. 2024, sec. 3.3.1).

Parameter decompositions may allow narrower edits. adVersarial Parameter Decomposition (VPD) approximates each weight matrix by a sum of rank-one *subcomponents*, trained so that few are needed on any input (Bushnaq et al. 2026, sec. 2). It follows earlier attribution-based and stochastic decompositions (Braun et al. 2025; Bushnaq et al. 2025). If a subcomponent is specific to one language, changing its scalar coefficient might alter that language while sparing its relatives.[1]

Bushnaq tested this possibility by training scalar coefficients of selected VPD subcomponents to suppress German (Bushnaq 2026). One subcomponent, chosen by its interpretation label, reached chance-level German with little English damage and spared French and Spanish, although it also damaged Italian.

Here I extend that test to three pairs of related languages: German (de)–Dutch (nl), Spanish (es)–Portuguese (pt) and French (fr)–Italian (it). In each pair, the first language is the target of deliberate damage and the second measures selectivity. A fixed numerical rule selects one subcomponent from target and English text. I compare its edits with rank-one LoRA, full-matrix fine-tuning and neuron edits. The related language in each pair (labelled *protected* in tables and figures) is excluded from the selection and training inputs in this comparison.

An edit could appear more selective simply because it changes the target less. I therefore calibrate every method to a target cross-entropy rise of 0.5 or 1 nat/token on validation text. I then evaluate the fixed edits on new European Parliament session text (Koehn 2005) to test whether their relative selectivity persists.

[1] In the VPD report's emoticon-editing experiment, a LoRA update trained on 947 examples outperformed a manual write-vector edit on both off-target KL measures (Bushnaq et al. 2026, sec. 6).

2 Methods

2.1 Model and data

The base model W_0 is Goodfire's four-layer, 67M-parameter decoder-only model, trained on a subset of the Pile and decomposed in the VPD report (Bushnaq et al. 2026, sec. 3.1; Gao et al. 2020). Its 24 weight matrices comprise the query, key, value and output projections of attention and two MLP matrices per layer. The decomposition supplies 38,912 rank-one subcomponents, $A_c = u_c v_c^\top$, which together form the VPD dictionary.

Europarl supplies one window per session date, starting at an aligned speaker turn so that languages share a common source context. Training windows serve selection and learning, and validation windows serve calibration. Evaluation uses 64 development dates and then 58 further dates, the confirmation set. I measure the change in mean next-token cross-entropy (CE) from W_0, in nat/token. To check sensitivity to the chosen target-training subset, I repeat selection and training on three overlapping subsets, termed draws. Learned runs also use different seeds. The English retain pool combines Europarl, WikiText-2 (Merity et al. 2017) and English-filtered Pile text and is fixed across draws. Data sizes and implementation details are in Appendix A.

2.2 Subcomponent selection

An edit changes one weight matrix by $W_m \leftarrow W_m + \alpha A_c$, leaving the other subcomponents and the decomposition residual fixed. The subcomponent is not normalized. In VPD each subcomponent has a scalar mask coefficient equal to 1 in the unedited model; the edit changes it to $1 + \alpha$, and the edit norm is $|\alpha| \, \|A_c\|_F$.

The selection rule ranks subcomponents by the target CE gradient relative to English Fisher curvature. The score favors target-CE increases with low predicted English KL for small coefficient changes:

$$\frac{g_c^2}{q_c + \varepsilon \|A_c\|_F^2},$$

where g_c is the derivative of target CE with respect to the mask coefficient, q_c is the directional Fisher estimate on the English retain pool, and ε supplies a small curvature floor. Because this score is local, the ranked shortlist is then tested at finite doses on target training text. The rule uses $\alpha = \text{sign}(g_c)\, a$, increasing the dose a up to 2. It accepts the first subcomponent to reach a target CE rise of 1 nat/token, falling back to the first that reaches 0.5 if none reaches 1. If neither threshold is reached, no edit is selected.

2.3 Comparison methods

- **Rank-one LoRA** (Hu et al. 2022) learns both update vectors in layer 0's attention output matrix, where all three selected subcomponents lie, matching their matrix and rank while learning its own direction.
- **Full-matrix fine-tuning** allows learned changes across all 24 weight matrices.
- **Neuron edits** select the top k MLP neurons by their excess activation frequency on target text over English, then reduce their output weights. Both k and the scale are calibrated to the target effect (Tang et al. 2024; Kojima et al. 2024).
- **Randomized-dictionary edits** apply the subcomponent selection rule after independent random signed permutations of the dictionary's input and output coordinates in each matrix. This preserves subcomponent norms and pairwise angles within each matrix while changing their alignment with the model's weights. The control was planned to test whether disrupting this alignment changes selectivity under the same rule, provided matched edits could be found.

The learned updates are trained for 200 steps per draw, minimizing a per-window hinge penalty for target CE rises below 1 nat/token plus English KL divergence from W_0, with weight 1. The final update is scaled to the required target effect.

2.4 Matching target effects

I calibrate every edit on target validation text to a CE rise of $\tau = 0.5$ or 1 nat/token, within 0.02. Only these *matched* edits enter the comparison. For the main comparison, a prespecified tolerance of $\tau \pm 0.1$ nat/token applies to each edit's mean target rise on evaluation text. I report the achieved target effects alongside related-language damage.

For each target, the selected subcomponent and calibrated doses were unchanged across the three draws, leaving six distinct edits across targets and effect sizes. All selected subcomponent edits use the negative direction, so their coefficient is $1 - a$: $a = 1$ removes the contribution and $a > 1$ reverses it. None is an exact ablation: three reduce the contribution without changing its sign and three reverse it, two only slightly (Table 1).

2.5 Evaluation

The related languages were excluded from this study's selection and training inputs, but my earlier experiments had used their development text and identified the German subcomponent with Dutch in view. The edits, comparators and main analysis were fixed before the development readout on 64 session dates. The unchanged edits were then evaluated once on 58 further session dates, without recalibration. These new dates test the fixed edits on new text in the same languages, leaving the selection rule's generalization untested. Appendix A.3 records the prior access and analysis chronology.

Edits were eligible only if no English evaluation set had a CE rise above 0.1 nat/token. I compare related-language CE rises using simultaneous 95% bootstrap intervals over session dates. These intervals condition on the selected edits and training runs. The analysis details are in Appendix A.2.

3 Results

3.1 Related-language damage

On the 58 new session dates, the six subcomponent edits raised related-language CE by 0.0003–0.0053 nat/token, compared with 0.027–0.876 across individual comparator runs (Figure 1; Table 2). All matched edits passed the English eligibility checks. Every edit in the main comparison also stayed within the prespecified target-effect tolerance.

All 18 mean differences between subcomponent edits and the three comparators have simultaneous 95% intervals below zero, with upper bounds at -0.023 nat/token or lower (Figure 2, top row; Appendix B). These comparisons reuse six subcomponent edits, each compared with three methods. The related-language advantage thus persists on text not used to calibrate the edits.

A separate post hoc analysis of related-language KL divergence from the base model gave the same ordering, with all 18 intervals below zero.

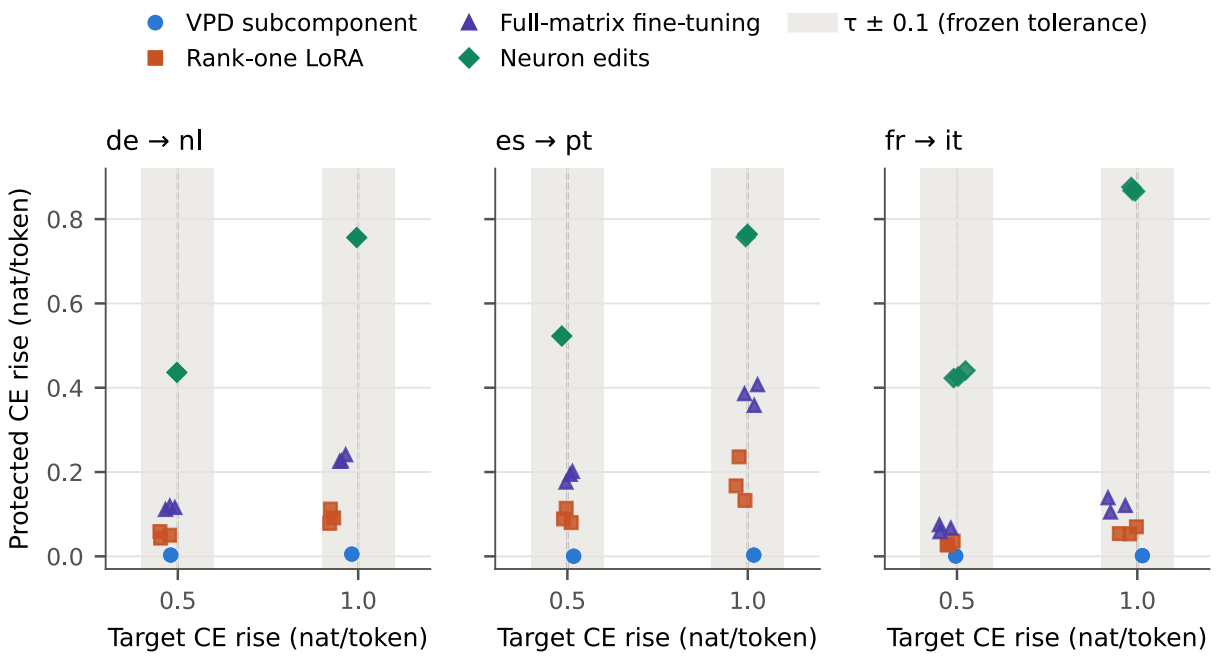


Figure 1: Target and protected-language CE rises on the 58 new dates. Each subcomponent edit is shared by all three draws; the other methods have one edit per draw. Shading marks the prespecified evaluation tolerance $\tau \pm 0.1$ nat/token.

Table 1: Selected subcomponent edits, shared across training draws. Coefficient: $1 - a$; edit norm: $|\alpha| \cdot \|A\|$. CE rises are in nat/token. C: confirmation; D: development.

Pair	Subcomponent	τ	Dose a	Coefficient	Edit norm	Target rise, D	Target rise, C	Protected rise, C
de → nl	0.attn.o:268	0.5	0.681	0.319	0.29	0.49	0.48	0.0030
		1	0.945	0.055	0.40	1.00	0.98	0.0053
es → pt	0.attn.o:676	0.5	1.026	−0.026	0.34	0.53	0.52	0.0003
		1	1.388	−0.388	0.47	1.05	1.02	0.0030
fr → it	0.attn.o:640	0.5	0.762	0.238	0.31	0.53	0.50	0.0008
		1	1.039	−0.039	0.42	1.06	1.01	0.0017

3.2 Achieved target effects

Staying within $\tau \pm 0.1$ still allows two methods' target effects to differ, so a post hoc check compares them directly to assess whether the related-language advantage came with a smaller target effect. The simultaneous interval for the between-method target gap lies wholly within ±0.1 nat/token in 15 of 18 comparisons (Figure 2, bottom row). This agreement band was chosen after the results. The three remaining gaps occur at $\tau = 1$ and have positive means: the subcomponent edit changes the target more on average, although one interval also permits a smaller effect. The observed advantage therefore does not appear to be driven by a weaker target effect. The subcomponent edits also had larger Frobenius weight-change norms than the learned updates (Table 2).

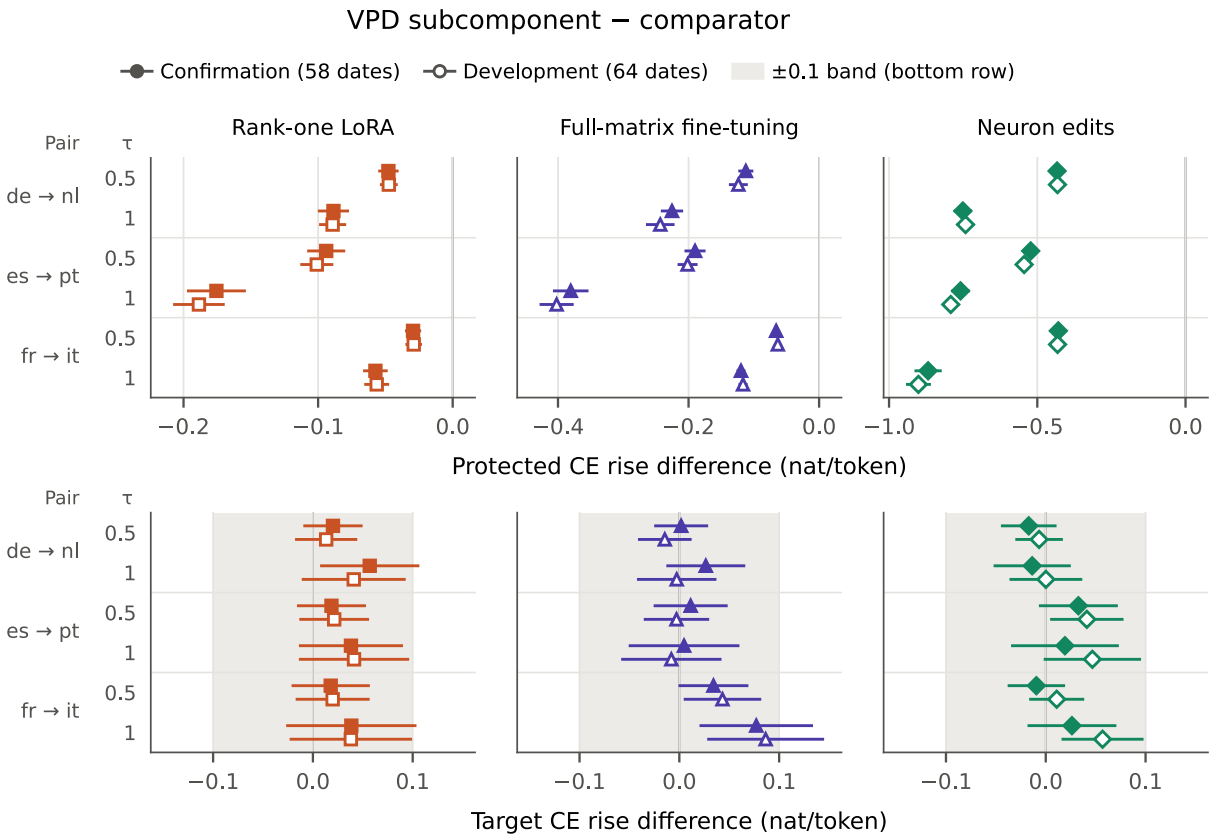


Figure 2: Differences in CE rise, subcomponent minus comparator, with simultaneous 95% intervals. Filled markers: confirmation; open markers: development. Top: the main protected-language contrasts, with separate axis scales by comparator. Bottom: achieved target-effect differences from the post hoc analysis, with its ±0.1 nat/token agreement band. Negative values favour the subcomponent edit in the top row; positive values indicate a larger target effect in the bottom row.

3.3 Preservation of English predictions

Passing the English CE limit does not imply equal preservation of English predictions. In a post hoc comparison of $D_{\mathrm{KL}}(p_{W_0} \,\|\, p_{\mathrm{edit}})$, LoRA had lower English KL than the subcomponent edits in 72/72 development comparisons and full-matrix fine-tuning in 65/72, across targets, draws, τ values and four English sets. On the single English set from the new dates, each learned method had lower KL in 18/18 comparisons across targets, draws and τ values (Table 2). These are dependent descriptive counts that reuse edits and text sets.

On the new dates, the subcomponent edits' English KL was nevertheless below 10^{-3} nat/token. Both learned methods explicitly optimize English KL, whereas none of the methods optimizes retention of the related language.

Table 2: Confirmation effects on 58 dates and development English KL. Each method–τ row covers 3 targets × 3 draws; VPD repeats one fixed edit per target. Confirmation ranges span these 9 run entries. Development English KL spans all four English sets per run: the validation set and three evaluation sets. CE rises and KL are in nat/token. C: confirmation; D: development. KL is from the base model; edit norm is the Frobenius norm of the weight change.

Method	τ	Target rise	Protected rise	English rise	English KL, C (10^{-3})	English KL, D (10^{-3})	Edit norm
VPD subcomponent	0.5	0.48–0.52	0.0003–0.0030	0.0002–0.0012	0.28–0.41	0.17–0.56	0.29–0.34
VPD subcomponent	1	0.98–1.02	0.0017–0.0053	0.0003–0.0019	0.49–0.77	0.31–1.01	0.40–0.47

Method	τ	Target rise	Protected rise	English rise	English KL, C (10^{-3})	English KL, D (10^{-3})	Edit norm
Rank-one LoRA	0.5	0.45–0.51	0.027–0.114	−0.0002 to 0.0001	0.05–0.08	0.04–0.16	0.12–0.17
Rank-one LoRA	1	0.92–1.00	0.053–0.236	−0.0004 to 0.0001	0.08–0.15	0.08–0.25	0.15–0.24
Full-matrix fine-tuning	0.5	0.45–0.51	0.058–0.201	0.0007–0.0014	0.08–0.21	0.08–0.24	0.07–0.08
Full-matrix fine-tuning	1	0.92–1.03	0.105–0.407	0.0010–0.0020	0.16–0.42	0.15–0.49	0.10–0.11
Neuron edits	0.5	0.48–0.52	0.423–0.523	−0.0008 to 0.0010	3.29–5.63	0.61–5.45	1.12–3.16
Neuron edits	1	0.98–1.00	0.756–0.876	0.013–0.038	13.38–38.67	11.85–39.86	5.85–9.80

3.4 Expanded training comparison

The main learned updates used 200 steps. Their selected learning rates lay at the edges of the tested grids (upper for LoRA, lower for full-matrix fine-tuning), so the grids may have missed better settings. To examine sensitivity to this recipe, an exploratory comparison used 2,000 steps, a larger hinge margin, wider learning-rate grids and more seeds, then calibrated the updates to the same τ. Evaluation used the previously evaluated development dates.

This expanded recipe reduced related-language damage for full-matrix fine-tuning in all six target and τ combinations and for LoRA in five, but the subcomponent edits retained their advantage. Achieved target effects differed from the subcomponent edits by at most 0.07 nat/token. Several training settings changed together, so the improvement cannot be attributed to step count alone. The training runs were not shown to converge at either budget, so the comparison does not establish the best selectivity these learned methods could attain. Appendix A.1 gives the extended comparison methods.

3.5 Choice within the VPD dictionary

Two alternative rules pick the subcomponent whose unit-norm direction has the largest absolute projection of a learned update. A third uses the target-minus-English difference in VPD's learned scores for how much of each subcomponent to retain, termed causal importance (Appendix A.1). In these descriptive comparisons, all three rules caused at least as much related-language damage as the main selector wherever estimates were available. Some alternatives had no matched edit, and some comparisons used fewer draws. The causal-importance rule raised related-language CE by 0.48–1.37 nat/token more than the main selector on the new dates, at similar target effects. Reusing the previously chosen German subcomponent produced the same edit as the main selector. These comparisons show that the choice of direction matters for the tested edits.

3.6 Contribution of the dictionary

Testing what the dictionary contributes requires comparisons with directions outside it. The planned randomized-dictionary control produced no matched edit in any target–draw combination: no shortlisted subcomponent reached even 0.5 nat/token of target damage within the dose limit of 2. It therefore did not compare selectivity at the required target effect. Larger doses remain untested.

In exploratory comparisons, a whitened rank-one selector searched outside the dictionary using the target gradient and an approximate English Fisher curvature. Its calibrated edits caused more related-language damage than the VPD edits on development text. However, it used the selected VPD subcomponent's norm and the main selector's curvature floor, so this comparison does not isolate the dictionary's contribution. Applying the main selector to singular components of the weight matrices yielded no eligible comparison: every edit that reached τ exceeded the English CE limit. These controls leave the dictionary's contribution unresolved.

4 Discussion

The methods rank differently for English and for the related languages. Learned updates better preserve English predictions, which their objective explicitly protects, while the selected VPD directions better spare the related languages absent from that objective. The comparison does not test learned updates trained to preserve the related language itself. Calibration also matches only the mean target effect. Because the learned objective penalizes each target window separately, equal means may hide different spreads of target damage across windows.

These conclusions concern six fixed edits in one small model, across three language pairs and two validation-calibrated target levels. They describe next-token prediction on parliamentary text.

Prior use of related-language development text and the earlier choice of the German subcomponent (Section 2.5), together with the inconclusive dictionary controls, limit what can be attributed to the selection procedure or decomposition. The dictionary's contribution and the rule's reliability on new tasks remain unresolved, and the study does not identify a language-specific mechanism. Within these limits, six fixed subcomponent edits in this model damaged the related languages less than rank-one LoRA, full-matrix fine-tuning and neuron edits calibrated to the same validation target effect, while the learned updates preserved English better.

Code and data. GitHub provides code, saved measurements and reproduction instructions. Analysis can be replayed from the per-date measurements; the confirmatory text and token windows are not released, preventing a rerun of that model evaluation from this checkout.

AI assistance. I used OpenAI Codex and Anthropic Claude for research planning, code development, data analysis and manuscript preparation. Scientific Agent Skills (Kassis et al. 2026) supplied the scientific-writing, scientific-critical-thinking, peer-review and citation-management workflows used to review organization, claims and references.

References

Braun, Dan, Lucius Bushnaq, Stefan Heimersheim, Jake Mendel, and Lee Sharkey. 2025. *Interpretability in Parameter Space: Minimizing Mechanistic Description Length with Attribution-Based Parameter Decomposition*. https://doi.org/10.48550/arXiv.2501.14926.

Bushnaq, Lucius. 2026. *Exploration: Fine-Tuning with Parameter Decomposition*. LessWrong, 25 June 2026. https://www.lesswrong.com/posts/ieoWstubDQWLrMnhH/exploration-fine-tuning-with-parameter-decomposition.

Bushnaq, Lucius, Dan Braun, Oliver Clive-Griffin, et al. 2026. *Interpreting Language Model Parameters*. Goodfire and MATS. https://www.goodfire.com/research/interpreting-lm-parameters.

Bushnaq, Lucius, Dan Braun, and Lee Sharkey. 2025. *Stochastic Parameter Decomposition*. https://doi.org/10.48550/arXiv.2506.20790.

Efron, Bradley. 1979. "Bootstrap Methods: Another Look at the Jackknife." *The Annals of Statistics* 7 (1): 1–26. https://doi.org/10.1214/aos/1176344552.

Federmann, Christian, Tom Kocmi, and Ying Xin. 2022. "NTREX-128 – News Test References for MT Evaluation of 128 Languages." *Proceedings of the First Workshop on Scaling up Multilingual Evaluation* (Online), 21–24. https://doi.org/10.18653/v1/2022.sumeval-1.4.

Gao, Leo, Stella Biderman, Sid Black, et al. 2020. *The Pile: An 800GB Dataset of Diverse Text for Language Modeling*. https://doi.org/10.48550/arXiv.2101.00027.

Hu, Edward J., Yelong Shen, Phillip Wallis, et al. 2022. "LoRA: Low-Rank Adaptation of Large Language Models." *International Conference on Learning Representations*. https://openreview.net/forum?id=nZeVKeeFYf9.

Kassis, Timothy, Vinayak Agarwal, Yuhuan He, Darshil Patel, and Aubrey M. Brueckner. 2026. *Scientific Agent Skills: A Library of Procedural Knowledge for Research Agents*. https://doi.org/10.48550/arXiv.2609.00065.

Kingma, Diederik P., and Jimmy Ba. 2015. "Adam: A Method for Stochastic Optimization." *International Conference on Learning Representations*. https://arxiv.org/abs/1412.6980.

Koehn, Philipp. 2005. "Europarl: A Parallel Corpus for Statistical Machine Translation." *Proceedings of Machine Translation Summit X: Papers* (Phuket, Thailand), 79–86. https://aclanthology.org/2005.mtsummit-papers.11/.

Kojima, Takeshi, Itsuki Okimura, Yusuke Iwasawa, Hitomi Yanaka, and Yutaka Matsuo. 2024. "On the Multilingual Ability of Decoder-Based Pre-Trained Language Models: Finding and Controlling Language-Specific Neurons." *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)* (Mexico City, Mexico), 6919–71. https://doi.org/10.18653/v1/2024.naacl-long.384.

Kunstner, Frederik, Philipp Hennig, and Lukas Balles. 2019. "Limitations of the Empirical Fisher Approximation for Natural Gradient Descent." *Advances in Neural Information Processing Systems* 32: 4158–69. https://proceedings.neurips.cc/paper/2019/hash/46a558d97954d0692411c861cf78ef79-Abstract.html.

Mallat, Stéphane G., and Zhifeng Zhang. 1993. "Matching Pursuits with Time-Frequency Dictionaries." *IEEE Transactions on Signal Processing* 41 (12): 3397–415. https://doi.org/10.1109/78.258082.

Martens, James. 2020. "New Insights and Perspectives on the Natural Gradient Method." *Journal of Machine Learning Research* 21 (146): 1–76. https://jmlr.org/papers/v21/17-678.html.

Martens, James, and Roger Grosse. 2015. "Optimizing Neural Networks with Kronecker-Factored Approximate Curvature." *Proceedings of the 32nd International Conference on Machine Learning* 37: 2408–17. https://proceedings.mlr.press/v37/martens15.html.

Meng, Kevin, David Bau, Alex Andonian, and Yonatan Belinkov. 2022. "Locating and Editing Factual Associations in GPT." *Advances in Neural Information Processing Systems* 35: 17359–72. https://doi.org/10.52202/068431-1262.

Merity, Stephen, Caiming Xiong, James Bradbury, and Richard Socher. 2017. "Pointer Sentinel Mixture Models." *International Conference on Learning Representations*. https://arxiv.org/abs/1609.07843.

Montiel Olea, José Luis, and Mikkel Plagborg-Møller. 2019. "Simultaneous Confidence Bands: Theory, Implementation, and an Application to SVARs." *Journal of Applied Econometrics* 34 (1): 1–17. https://doi.org/10.1002/jae.2656.

Pati, Y. C., R. Rezaiifar, and P. S. Krishnaprasad. 1993. "Orthogonal Matching Pursuit: Recursive Function Approximation with Applications to Wavelet Decomposition." *Proceedings of 27th Asilomar Conference on Signals, Systems and Computers* 1: 40–44. https://doi.org/10.1109/ACSSC.1993.342465.

Tang, Tianyi, Wenyang Luo, Haoyang Huang, et al. 2024. "Language-Specific Neurons: The Key to Multilingual Capabilities in Large Language Models." *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (Bangkok, Thailand), 5701–15. https://doi.org/10.18653/v1/2024.acl-long.309.

Yao, Yunzhi, Peng Wang, Bozhong Tian, et al. 2023. "Editing Large Language Models: Problems, Methods, and Opportunities." *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing* (Singapore), 10222–40. https://doi.org/10.18653/v1/2023.emnlp-main.632.

Zhang, Zhihao, Jun Zhao, Qi Zhang, Tao Gui, and Xuanjing Huang. 2024. "Unveiling Linguistic Regions in Large Language Models." *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (Bangkok, Thailand), 6228–47. https://doi.org/10.18653/v1/2024.acl-long.338.

Appendix A — Method details

Data. Europarl v7 (Koehn 2005) supplies one 257-token window per session date (256 predictions). Windows start at the same aligned speaker turn in each language, but each language is tokenized and truncated independently, so the windows need not cover identical content. Each target has three draws of 64 of its 128 training windows and 32 validation windows for calibration. The English validation set has 64 windows: 32 Europarl and 32 English-filtered Pile. The retain pool has 256 windows: 128 Europarl English training, 64 WikiText-2 validation and 64 English-filtered Pile validation.

Selection. The model Fisher estimate q_c averages squared directional scores using eight model-sampled label sets per retain window (Martens 2020; Kunstner et al. 2019). The floor ε is the 1st percentile of $q_c/\|A_c\|_F^2$. The top 64 subcomponents are re-ranked using the larger of q_c and a second estimate on 256 other English windows, before the dose search of Section 2.2. Curvature checks passed on the NVIDIA L40S used for the original runs, but the batch-splitting check failed on the tested A10 and L4. Verified selector reruns cover only the German target's first draw on the L40S, with the VPD dictionary and one randomized dictionary. GPU-dependent Fisher-label sampling can change selections on other hardware.

Calibration. The first dose-grid crossing of τ is bracketed, then bisected for at most five steps until within the matching tolerance. Neuron edits scale the top k neurons' output weights by $1 - f$: calibration takes the

smallest power-of-two k from 1 to 256 whose full removal reaches τ, then calibrates f. Randomized dictionaries use a separate seed per draw.

Learned updates. The 200-step updates use Adam (Kingma and Ba 2015), 10 warm-up steps, gradient clipping at 1.0, and 8 target and 8 English windows per step. Calibration scales the final update D to βD, with $\beta \leq 1$. The learning rate is chosen once per target on draw 0, by the lowest English validation KL at $\tau = 1$. Every target chose 3e-3 for LoRA and 1e-6 for full-matrix fine-tuning, respectively the upper and lower edges of the tested grids.

A.1 Alternative selectors and extended training

Two within-dictionary alternatives use the first step of orthogonal matching pursuit (Pati et al. 1993; Mallat and Zhang 1993): they normalize each subcomponent to unit Frobenius norm, select the one with the largest absolute inner product with a learned update, and use that inner product's sign. One uses full-matrix fine-tuning; the other uses rank-one LoRA across all 24 matrices. VPD's learned causal-importance score sets the lower bound on a subcomponent's mask during decomposition training. A third rule selects the largest mean target-minus-English score with a negative sign, adapting Bushnaq's top-16 ranking to one subcomponent (Bushnaq 2026). The fourth alternative reuses the previously chosen German subcomponent. All use the main calibration rule. Some fail to reach τ or lack an eligible learned checkpoint; their contrasts include only the available matched, usable draws and use a separate simultaneous family.

The extended learned updates use 2,000 steps, a hinge margin of 1.5, wider learning-rate grids and three seeds per draw. The whitened selector maximizes the squared target gradient relative to damped K-FAC curvature on English text (Martens and Grosse 2015), then tests doses from 1/64 to 8. The SVD control applies the main selection rule unchanged to the singular components of all 24 weight matrices; these reconstruct the weights exactly, whereas VPD leaves a residual.

Extended contrasts compare unpaired means over each method's matched, usable runs. A sensitivity analysis excludes 7 of 27 LoRA runs whose English validation KL at step 2,000 exceeded twice their minimum from step 100 onward. Both analyses favour the subcomponent edits. Training diagnostics did not establish objective convergence and were not used to select edits.

These exploratory comparisons were specified before the runs. Analysis rules were fixed before readout, after selection outcomes and target and English validation effects were known. This chronology comes from the author's records, without independent timestamps.

A.2 Evaluation and intervals

Evaluation sets. Development uses 64 session dates. English eligibility checks use Europarl test text (64 windows), WikiText-2 test text (32) and Pile test documents (512). Confirmation uses 58 further dates (48 main and 10 reserve, analysed together), including English. An edit must pass both development and confirmatory English checks to enter a confirmatory comparison.

Main comparisons. The planned family contains four comparators × three language pairs × two values of τ. Each contrast is the subcomponent edit's CE rise minus the comparator's, averaged over matched, eligible draws and then over dates. Simultaneous 95% intervals use 2,000 shared bootstrap resamples of session dates (Efron 1979) and a max-|t| critical value (Montiel Olea and Plagborg-Møller 2019, supplement, Algorithm 3). Training draws and seeds, selected subcomponents, language pairs and the model remain fixed. The randomized-dictionary contrasts have no estimates because the control produced no feasible edits.

Target-effect checks. The prespecified tolerance for each edit's target CE rise was $\tau \pm 0.1$ nat/token; all edits in the main confirmation comparison met it. The post hoc agreement analysis instead compares the target effects of two methods. Its simultaneous family includes 54 coordinates per readout: target gaps, related-language CE contrasts and related-language KL contrasts, using 20,000 shared date resamples. KL is $D_{\mathrm{KL}}(p_{W_0} \,\|\, p_{\mathrm{edit}})$, averaged over next-token positions.

A.3 Prior data access

The selection, comparison and analysis rules were fixed on 11 October 2026 before the development readout. Later that day, the unchanged edits were evaluated once on 58 further dates under an amendment fixed

before those windows were parsed. The chronology relies on the author's records; there is no independently timestamped preregistration.

In an earlier experiment, I had chosen German subcomponent `0.attn.o:268` with Dutch in view. It was also the top German candidate in a pilot of this selector. Dutch, Portuguese and Italian development text and NTREX Dutch and Italian text (Federmann et al. 2022) had been read or used previously. The development dates therefore do not provide a fresh test. The Pile also contains a European Parliament component (Gao et al. 2020, sec. 2.17); overlap with the model's pretraining subset is unknown.

Appendix B — Comparison estimates

Table 3 gives the estimates and intervals plotted in Figure 2.

Table 3: VPD subcomponent minus comparator, in nat/token, with simultaneous 95% intervals. Protected: prespecified differences in protected-language CE rise. Target gap: differences in achieved target CE rise from the separate post hoc analysis.

Pair	τ	Comparator	Protected, confirmation	Protected, development	Target gap, confirmation
de → nl	0.5	Rank-one LoRA	−0.048 [−0.055, −0.040]	−0.047 [−0.054, −0.041]	+0.020 [−0.010, +0.050]
		Full-matrix fine-tuning	−0.112 [−0.124, −0.101]	−0.124 [−0.138, −0.109]	+0.002 [−0.025, +0.029]
		Neuron edits	−0.433 [−0.457, −0.410]	−0.432 [−0.456, −0.408]	−0.017 [−0.045, +0.011]
	1	Rank-one LoRA	−0.089 [−0.100, −0.077]	−0.089 [−0.099, −0.079]	+0.057 [+0.007, +0.107]
		Full-matrix fine-tuning	−0.225 [−0.242, −0.208]	−0.243 [−0.265, −0.221]	+0.027 [−0.013, +0.066]
		Neuron edits	−0.751 [−0.782, −0.720]	−0.742 [−0.772, −0.712]	−0.014 [−0.052, +0.025]
es → pt	0.5	Rank-one LoRA	−0.094 [−0.108, −0.080]	−0.101 [−0.113, −0.089]	+0.019 [−0.016, +0.053]
		Full-matrix fine-tuning	−0.190 [−0.206, −0.174]	−0.202 [−0.217, −0.186]	+0.011 [−0.026, +0.048]
		Neuron edits	−0.522 [−0.549, −0.496]	−0.544 [−0.570, −0.519]	+0.033 [−0.007, +0.072]
	1	Rank-one LoRA	−0.176 [−0.197, −0.154]	−0.189 [−0.208, −0.169]	+0.038 [−0.014, +0.090]
		Full-matrix fine-tuning	−0.380 [−0.408, −0.353]	−0.402 [−0.428, −0.376]	+0.005 [−0.051, +0.060]
		Neuron edits	−0.759 [−0.790, −0.728]	−0.792 [−0.821, −0.763]	+0.019 [−0.035, +0.073]
fr → it	0.5	Rank-one LoRA	−0.029 [−0.035, −0.023]	−0.029 [−0.035, −0.023]	+0.018 [−0.021, +0.057]
		Full-matrix fine-tuning	−0.066 [−0.071, −0.061]	−0.063 [−0.068, −0.058]	+0.034 [−0.001, +0.069]
		Neuron edits	−0.429 [−0.456, −0.402]	−0.432 [−0.456, −0.407]	−0.009 [−0.038, +0.019]
	1	Rank-one LoRA	−0.057 [−0.067, −0.048]	−0.056 [−0.066, −0.047]	+0.038 [−0.027, +0.104]
		Full-matrix fine-tuning	−0.120 [−0.126, −0.113]	−0.117 [−0.123, −0.110]	+0.077 [+0.020, +0.134]
		Neuron edits	−0.868 [−0.914, −0.822]	−0.901 [−0.943, −0.859]	+0.026 [−0.018, +0.071]